

Notes: Hart & Moore (JPE, 1990)

Property Rights and the Nature of the Firm

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1 Big picture

- **Question.** What determines the boundaries of the firm when contracts are incomplete and ownership affects the incentives of both owner-managers and employees?
- **Core definition of the firm.** A firm is identified with the collection of assets controlled by its owner. Ownership gives residual rights of control, especially the right to exclude others from the use of assets.
- **Central mechanism.** Because noncontractible human-capital investments are made before ex post bargaining, agents underinvest. Ownership changes who controls access to productive assets, which changes each agent's bargaining position and therefore ex ante investment incentives.
- **Employees matter.** The paper extends Grossman and Hart by explicitly considering employees/nonowners of assets, not only owner-managers.
- **Economic takeaway.** Firm boundaries trade off the benefits of coordinating complementary assets against the incentive costs of centralizing control over economically independent assets.

2 Incremental contribution

- The paper gives a formal incomplete-contracting answer to Coase's question about why some transactions occur inside firms and others through markets.
- It extends Grossman–Hart by specializing control as exclusion rights and by adding multiasset, multi-individual organizations with employees.
- It shows how control over physical assets indirectly creates control over human capital.
- It derives general rules for which assets should be grouped, separated, or jointly controlled.
- It accommodates partnerships, majority voting, worker cooperatives, consumer cooperatives, stochastic control, and more general governance structures.
- It provides a foundation for property-rights theories of vertical integration, organizational boundaries, asset ownership, authority, and incomplete contracts.

3 Model objects and measurement

3.1 Agents, assets, and timing

There is a finite set of agents $S = \{1, \dots, I\}$ and a finite set of assets $A = \{a_1, \dots, a_N\}$. At date 0, agents choose noncontractible human-capital investments x_i . At date 1, production and trade occur, and ex post bargaining divides the surplus.

Detailed interpretation. The timing isolates the hold-up problem: investments are chosen before bargaining, but contracts cannot commit the later division of surplus.

3.2 Coalition value and assumptions

If coalition S controls assets A , it produces value $v(S, A \mid x)$. The model assumes concavity, investment complementarity, superadditivity, and monotonic marginal returns in larger coalitions/assets.

Detailed interpretation. These assumptions make the model an underinvestment model: agents privately capture only part of the social marginal return to their investments.

3.3 Control structure

A control structure α maps every coalition S to the set of assets $\alpha(S)$ that S can control. Sole ownership is a special case: asset a belongs to $\alpha(S)$ if and only if the owner is in S .

Detailed interpretation. Ownership is represented as an access rule over coalitions. This is broader than simple title ownership.

4 Models and theoretical specifications

4.1 First-best investment

$$W(x) = V(x) - \sum_i C_i(x_i), \quad V_i(x^*) = C'_i(x_i^*) \quad \forall i.$$

Detailed interpretation. The first-best equates each agent's social marginal product to marginal investment cost.

4.2 Shapley payoff and private investment

$$B_i(\alpha \mid x) = \sum_{S \ni i} p(S) [v(S, \alpha(S) \mid x) - v(S \setminus \{i\}, \alpha(S \setminus \{i\}) \mid x)].$$

Agent i chooses x_i to maximize $B_i(\alpha \mid x) - C_i(x_i)$.

Detailed interpretation. The private marginal return is an average marginal contribution across coalitions, not the full social marginal product.

4.3 Main propositions

- **Proposition 1.** Every control structure generates underinvestment relative to first best.
- **Proposition 2.** If only one agent invests, that agent should own all assets.
- **Propositions 3–4.** Ambiguous control and multiple veto power are inefficient.
- **Propositions 5–6.** Idiosyncratic and indispensable agents should own the relevant assets.
- **Proposition 7.** If a group is key to an asset, simple majority voting among key agents is optimal.
- **Proposition 8.** Strictly complementary assets should be owned together.
- **Propositions 9–10.** Dispensable noninvestors and outside parties should not receive control; economically independent assets should remain independently controlled.
- **Propositions 11–12.** Essential assets and employment chains create additional constraints on efficient ownership patterns.

5 Theoretical design and identification logic

The paper identifies firm boundaries through how ownership changes date-1 bargaining positions and therefore date-0 investment incentives. It assumes efficient ex post bargaining, so inefficiency comes from investment incentives rather than failed trade. The model's main limitation is that it abstracts from financing, wealth constraints, risk aversion, hierarchy, and communication within the firm.

6 Tables and theoretical result blocks

6.1 Table 1: Forces Determining Whether 1 and 2 Should Integrate (1 Should Own 2)

Read: Table 1 summarizes the two-asset integration case in which party 1 may acquire asset a_2 from party 2. If agent 1's investment is important, integration tends to be desirable. If agent 2's investment is important, or if agent 2 is indispensable to a_2 , nonintegration tends to be desirable. Complementary assets push toward integration, while economically independent assets push toward nonintegration.

Detailed interpretation. Table 1 is not a regression table. It is a compact map of the model's comparative statics. Each row tells the reader which economic force pushes toward integration or nonintegration.

Economic message. Integrate when ownership strengthens important investments or coordinates complementary assets; do not integrate when it weakens indispensable or independent agents' incentives.

TABLE 1 FORCES DETERMINING WHETHER 1 AND 2 SHOULD INTEGRATE (1 Should Own 2)			
FORCE	TENDENCY TOWARD		PROPOSITION
	Integration	Nonintegration	
Agent 1's investment is important	x		2
Agent 2's investment is important		x	2
Agent 2 is indispensable to asset a_2		x	6
Asset a_2 is idiosyncratic to agent 2		x	5
Agent 1 is indispensable to asset a_2	x		6
Asset a_2 is idiosyncratic to agent 1	x		5
Assets a_1 and a_2 are complementary	x		8
Agent 2 is dispensable	x		9
Assets a_1 and a_2 are economically independent		x	10
Agent w 's investment is important	x		11

Table 1: Forces determining whether 1 and 2 should integrate.

7 Result map

- **Underinvestment mechanism.** Agents invest to improve their own Shapley payoff, not total surplus.
- **Indispensability mechanism.** An indispensable agent should own the asset because every productive coalition needs him anyway.
- **Complementarity mechanism.** Complementary assets should be jointly controlled because separate owners create multiple hold-up points.
- **Economic independence mechanism.** Independent assets should not be integrated because there is no synergy.
- **Employee incentive mechanism.** A worker cares about who controls the assets needed to make his human capital valuable.
- **Many-assets mechanism.** Common ownership of a bottleneck asset can be valuable when many downstream assets depend on it.

8 Short summary

- Hart and Moore provide a formal property-rights theory of firm boundaries.

- A firm is a collection of assets under common residual control.
- Ownership means the right to exclude others from assets, not necessarily the right to command people directly.
- Agents make noncontractible human-capital investments before ex post bargaining.
- Because investment returns are shared through bargaining, agents underinvest relative to first best.
- Complementary assets should be owned together, while economically independent assets should remain separately controlled.
- The paper explains both why firms integrate and why they stop integrating.

9 Conclusion

9.1 Core conclusion

Hart and Moore's central conclusion is that firm boundaries are determined by how asset ownership affects incentives for noncontractible human-capital investments.

9.2 Economic intuition

Physical assets shape bargaining power over human assets. If an employee's training is useful only with a particular machine, building, client list, or pipeline, then the owner of that asset becomes the employee's future bargaining partner.

9.3 Incremental contribution revisited

The paper moves beyond a buyer-supplier model by allowing many agents, many assets, employees, complementary assets, essential assets, outside parties, and alternative control structures.

9.4 Implications

Firms should integrate complementary assets, avoid integrating independent assets, and allocate control to indispensable agents when their presence is necessary for productive coalitions.

9.5 Limitations

The model assumes efficient ex post bargaining and abstracts from financing, hierarchy, wealth constraints, and communication within firms.

9.6 Takeaway

Ownership matters because it determines who must be bargained with after contracts run out. The efficient firm boundary best protects relationship-specific investments while avoiding unnecessary incentive dilution.